Notes: Vayanos & Vila (Econometrica, 2021)
A Preferred-Habitat Model of the Term Structure of Interest Rates

Contents

1 Big picture

- **Question.** How can one build a no-arbitrage term-structure model in which investors have preferences for specific maturities, arbitrage is limited, and central-bank policies such as forward guidance and quantitative easing have realistic effects on yields?
- **Core idea.** The term structure is determined by the interaction between preferred-habitat investors, who demand bonds of particular maturities, and risk-averse arbitrageurs, who intermediate across maturities through carry trades.
- **Main mechanism.** Shocks to the short rate and shocks to preferred-habitat demand are transmitted across maturities only because arbitrageurs choose to bear duration risk. When arbitrageurs are risk averse, they demand compensation for bearing this risk, so risk premia and yields are affected.
- **Key qualitative result.** Forward guidance affects long rates less than the expectations hypothesis predicts. Long rates underreact because arbitrageurs do not fully arbitrage away expected profits from carrying long bonds when short rates are expected to fall.
- **Demand-effect result.** With only the short rate as a risk factor, preferred-habitat demand shocks have global effects: demand changes at any maturity move all yields in the same direction with the same relative maturity pattern. With demand-risk factors, effects become more localized.
- **Risk-premium result.** Bond risk premia are positively related to the slope of the term structure because a steeper curve makes carry trades more attractive but also exposes arbitrageurs to short-rate risk.
- **Quantitative result.** In the calibration, lowering expected short rates over ten years by 100 bps lowers the 10-year yield by only about 35–50 bps. QE purchases equal to 12% of GDP lower the 10-year yield by roughly 25–30 bps and the 30-year yield by roughly 30–35 bps under the Fed QE1 maturity distribution.
- **Policy takeaway.** Forward guidance is strong at short maturities but weak at long maturities. QE is stronger at long maturities when purchases are tilted toward long bonds.

1.1 Incremental contribution of the paper

- **Formalization of preferred habitat.** Earlier preferred-habitat arguments were often informal or partial-equilibrium. This paper embeds maturity-clientele demand into a modern affine no-arbitrage term-structure model.
- **Limited-arbitrage bridge.** It explains how segmentation and no-arbitrage coexist: arbitrageurs connect maturity segments, but their risk aversion prevents full arbitrage.
- **Unified policy framework.** The model treats forward guidance and QE in the same equilibrium setting, allowing comparison of short-rate news versus bond-demand shocks.
- **Global versus local demand effects.** The paper clarifies when demand effects are global and when they become localized. Localization is not automatic; it requires additional demand-risk factors.
- **Quantitative calibration.** It calibrates the model to U.S. yield volatilities, yield-change correlations, trading-volume maturity shares, and demand-elasticity estimates, producing policy magnitudes that can be compared to QE and forward-guidance episodes.
- **Macro-finance link.** It converts a practitioner and central-bank narrative about preferred habitat into a tractable tool for studying risk premia, monetary policy transmission, and bond-market segmentation.

2 Data, calibration, and measurement

2.1 Term-structure data

The calibration uses U.S. government bond yield data from the Gurkaynak–Sack–Wright data sets. The main nominal-yield sample starts in November 1985, when all maturities from 1 to 30 years are available, and ends in January 2020. A real-yield calibration uses TIPS yields from January 1999 to January 2020. The paper works with end-of-month spot rates.

Interpretation: The model is theoretical, but the calibration is disciplined by the shape and dynamics of the U.S. yield curve. Matching yield levels is less central for policy experiments than matching yield volatilities, yield-change correlations, and the maturity distribution of trading volume.

2.2 Trading-volume data

Trading-volume moments come from the Federal Reserve 2004 data set, which reports primary-dealer trading volume in Treasury securities by maturity bucket. T-bills are excluded because they have special collateral and liquidity features, and TIPS volume is too small and not sufficiently split by maturity for the relevant period.

Interpretation: Trading volume identifies the maturity profile of preferred-habitat demand shocks. Without volume moments, long-maturity demand shocks and arbitrageur risk aversion would be much less separately disciplined.

2.3 Demand-elasticity calibration

The slope of preferred-habitat demand, α, is set using the Krishnamurthy–Vissing-Jorgensen estimate of how government-bond demand affects government-corporate spreads. The calibrated value in Table I is

$$\alpha = 5.21.$$

The paper treats this as an upper bound because outside returns for preferred-habitat investors may partly adjust when government yields change.

Interpretation: Yield and volume moments identify products such as $a\alpha$ and $a\theta$, not separately arbitrageur risk aversion a, demand slope α, and demand-shock scale θ. External evidence on demand elasticity is needed to pin down α.

2.4 Calibrated model objects

The calibration sets the maximum bond maturity to $T = 30$ years and assumes one demand factor independent of the short rate. Preferred-habitat demand is parameterized as

$$\alpha(\tau) = \alpha e^{-\delta_\alpha \tau}, \qquad \theta(\tau) = \theta \left(e^{-\delta_\alpha \tau} - e^{-\delta_\theta \tau} \right),$$

for $\tau < T$, and zero beyond T.

Interpretation: The exponential specification makes demand stronger at some maturities and declining at very long maturities. The finite 30-year horizon corresponds to the maximum maturity of U.S. Treasury bonds in the calibration.

3 Model primitives

3.1 Bonds, prices, and yields

Time is continuous. The term structure consists of zero-coupon government bonds with maturity $\tau \in (0, \infty)$. A bond with maturity τ pays one unit at $t + \tau$. Its price and yield are $P_t^{(\tau)}$ and $y_t^{(\tau)}$, with

$$y_t^{(\tau)} = -\frac{\log P_t^{(\tau)}}{\tau}.$$

The short rate r_t is the limit of $y_t^{(\tau)}$ as $\tau \to 0$.

3.2 Arbitrageurs

Arbitrageurs invest in bonds and in the short rate. If $X_t(\tau)d\tau$ is the present-value position in bonds with maturities in $[\tau, \tau + d\tau]$, their budget constraint is

$$dW_t = \left(W_t - \int_0^\infty X_t(\tau)d\tau \right) r_t dt + \int_0^\infty X_t(\tau) \frac{dP_t^{(\tau)}}{P_t^{(\tau)}} d\tau.$$

They maximize a mean–variance objective over instantaneous wealth changes:

$$\max_{\{X_t(\tau)\}_{\tau>0}} \left\{ E_t(dW_t) - \frac{a}{2} Var_t(dW_t) \right\}.$$

Interpretation: Arbitrageurs are the model's limited-arbitrage agents. The coefficient a is the key friction: when $a = 0$, arbitrage is risk neutral and the model converges toward expectations-hypothesis logic; when $a > 0$, arbitrageurs require risk premia.

3.3 Preferred-habitat investors

Investors with habitat τ hold only the bond of maturity τ. Their demand, expressed in present-value terms, is

$$Z_t(\tau) = -\alpha(\tau) \log P_t^{(\tau)} - \beta_t^{(\tau)}.$$

The demand intercept is

$$\beta_t^{(\tau)} = \theta_0(\tau) + \sum_{k=1}^{K} \theta_k(\tau) \beta_{k,t}.$$

The demand risk factors $\beta_{k,t}$ determine time-varying preferred-habitat demand. If $\theta_k(\tau)$ is flat, the shock is global; if it peaks near a maturity, the shock is local.

Interpretation: Preferred-habitat investors create segmentation. They do not arbitrage across maturities; instead, they create maturity-specific demand that arbitrageurs must absorb.

3.4 State-vector dynamics

The state vector is

$$q_t = (r_t, \beta_{1,t}, \ldots, \beta_{K,t})'.$$

It follows

$$dq_t = -\Gamma(q_t - rE)dt + \Sigma dB_t,$$

where $E = (1, 0, \ldots, 0)'$. This nests independent short-rate and demand factors, correlated shocks, and cross-factor mean reversion.

Interpretation: The model remains affine because state variables follow linear Gaussian dynamics and prices are exponential-affine in q_t.

4 Equilibrium and theoretical specifications

4.1 Affine bond prices

The equilibrium price of a bond of maturity τ takes the exponential-affine form

$$P_t^{(\tau)} = \exp\left[-A(\tau)'q_t - C(\tau)\right].$$

In the one-factor short-rate case, this becomes

$$P_t^{(\tau)} = \exp\left[-A_r(\tau)r_t - C(\tau)\right].$$

Interpretation: This puts the model inside the affine no-arbitrage family while giving the affine coefficients an equilibrium interpretation based on preferred-habitat demand and arbitrageur risk bearing.

4.2 Short-rate transmission and forward guidance

When the short rate falls, bonds become attractive relative to the short rate. Arbitrageurs borrow short and buy bonds, raising bond prices and lowering yields. But because this carry trade exposes them to future short-rate risk, they do not arbitrage fully.

Interpretation: This yields underreaction: longer yields respond less to future short-rate news than under the expectations hypothesis. The more risk averse arbitrageurs are, and the more price-elastic preferred-habitat demand is, the weaker the transmission from short to long rates.

4.3 Bond risk premia and slope predictability

The model generates a positive relationship between bond risk premia and the slope of the term structure. In the Fama–Bliss regression, expected excess bond returns are predicted by the forward-spot spread. In the Campbell–Shiller regression, yield changes underreact relative to the expectations hypothesis.

Interpretation: A steep term structure makes long-bond carry trades attractive. But because arbitrageurs require compensation for short-rate risk, the positive expected excess return is not fully arbitraged away.

4.4 Demand effects without demand risk

Suppose the demand intercept changes from $\theta_0(\tau)$ to $\theta_0(\tau) + \Delta\theta_0(\tau)$. In the one-factor model, demand effects depend on the scalar index

$$\int_0^\infty \Delta\theta_0(\tau)A_r(\tau)d\tau.$$

If this index is positive, all yields rise; if it is negative, all yields fall. Relative effects across maturities do not depend on the maturity at which the demand shock originated.

Interpretation: This is the paper's global demand-effect result. With only short-rate risk, arbitrageurs care about how a demand shock changes their aggregate short-rate exposure, not exactly where along the curve the shock originates.

4.5 Demand effects with demand risk

When preferred-habitat demand itself is stochastic, arbitrageurs face multiple risk factors. Demand shocks that originate at different maturities load differently on those factors. Effects become more localized: short-maturity demand shocks move short yields more, while long-maturity demand shocks move long yields more.

Interpretation: Localized preferred-habitat effects are not simply assumed; they arise from additional risk factors and limited risk-bearing capacity.

5 Identification and theoretical strategy

5.1 Theoretical identification

This is a structural theory and calibration paper, not a reduced-form causal paper. Identification comes from mapping model primitives to observable yield-curve moments. The key theoretical objects are:

- arbitrageur risk aversion a;
- preferred-habitat demand elasticity α;
- demand-shock size θ;
- short-rate and demand-factor mean reversion (κ_r, κ_β);
- maturity profiles $(\delta_\alpha, \delta_\theta)$.

5.2 Moment mapping

Table I maps model parameters to empirical moments. Short-rate volatility and mean reversion are tied to the volatility of the one-year yield. Demand-factor dynamics are tied to long-maturity yield volatility. Preferred-habitat demand slope is tied to correlations between short- and long-yield changes. Demand-profile decay parameters are tied to trading-volume shares at short and long maturities.

Interpretation: The identification strategy is a calibrated-moment strategy. The model is not estimated by full maximum likelihood; instead, parameters are chosen so that economically interpretable moments discipline economically interpretable primitives.

5.3 Policy-counterfactual identification

The policy experiments use the calibrated model to ask counterfactual questions:

- What happens to yields when expected future short rates change but preferred-habitat demand does not?
- What happens when preferred-habitat demand shifts because a central bank buys bonds of specific maturities?
- How do effects differ if purchases are concentrated at long maturities?

Interpretation: The policy analysis identifies model-implied impulse responses, not historical event-study treatment effects. Its credibility depends on whether the calibrated model captures the relevant term-structure moments and demand elasticities.

5.4 Main limitations

- The calibration assumes one demand factor independent of the short rate.
- Arbitrageur wealth and entry into arbitrage are fixed rather than endogenous.
- Preferred-habitat demand is reduced form, though an optimizing foundation is provided in the supplement.
- The model abstracts from broader macroeconomic feedback from the term structure to investment and output.
- The policy effects are model-implied counterfactuals, not direct high-frequency event-study estimates.

6 Tables and figures

6.1 Table I: Calibration of model parameters and Figure 1: Model-generated and empirical moments

Read: Table I reports the main nominal-yield calibration. The short-rate mean reversion is $\kappa_r = 0.125$, short-rate diffusion is $\sigma_r = 0.0146$, demand-factor mean reversion is $\kappa_\beta = 0.053$, the product $a\theta = 3155$, the product $a\alpha = 35.3$, and preferred-habitat demand slope is $\alpha = 5.21$. Figure 1 compares model-generated and empirical moments: yield volatilities, annual-change volatilities, yield-change correlations, first principal component, and Fama–Bliss and Campbell–Shiller regression coefficients.

Interpretation: The table explains what identifies the calibration: short-rate parameters from one-year yield moments, demand-factor parameters from longer-yield moments, and maturity-demand profiles from trading-volume shares. Figure 1 shows that the model fits the main factor structure and qualitative return-predictability patterns, though it underpredicts long-maturity predictability.

Economic message: The calibration is disciplined enough to evaluate policies, but not overfit: it matches core volatilities, correlations, and volume shares while leaving some return-predictability moments as out-of-sample checks.

TABLE I

CALIBRATION OF MODEL PARAMETERS FOR THE MAIN SAMPLE OF NOMINAL YIELDS

Parameter	Value	Empirical Moment	Value
κ_r Mean reversion of r_t	0.125	$\sqrt{\mathbb{V}\mathrm{ar}(y_t^{(1)})}$ Volatility 1-year yield – Levels	2.62
σ_r Diffusion of r_t	0.0146	$\sqrt{\mathbb{V}\mathrm{ar}(y_{t+1}^{(1)} - y_t^{(1)})}$ Volatility 1-year yield – Annual changes	1.27
κ_β Mean reversion of β_t	0.053	$\frac{1}{30}\sum_{\tau=1}^{30}\sqrt{\mathbb{V}\mathrm{ar}(y_t^{(\tau)})}$ Volatility τ-year yield – Levels, average over τ	2.20
$a\theta$ Arbitrageur risk aversion × Preferred habitat demand shock	3155	$\frac{1}{30}\sum_{\tau=1}^{30}\sqrt{\mathbb{V}\mathrm{ar}(y_{t+1}^{(\tau)} - y_t^{(\tau)})}$ Volatility τ-year yield – Annual changes, average over τ	0.796
$a\alpha$ Arbitrageur risk aversion × Preferred habitat demand slope	35.3	$\frac{1}{30}\sum_{\tau=1}^{30}\mathbb{C}\mathrm{orr}(y_{t+1}^{(1)} - y_t^{(1)}, y_{t+1}^{(\tau)} - y_t^{(\tau)})$ Correlation 1-year yield with τ-year yield – Annual changes, average over τ	0.504
δ_α Preferred habitat demand shock – short maturities	0.297	$\frac{\sum_{0<\tau\leq 2}\mathrm{Volume}(\tau)}{\sum_{0<\tau\leq 30}\mathrm{Volume}(\tau)}$ Relative volume for maturities $\tau \in (0, 2]$	0.199
δ_θ Preferred habitat demand shock – long maturities	0.307	$\frac{\sum_{11<\tau\leq 30}\mathrm{Volume}(\tau)}{\sum_{0<\tau\leq 30}\mathrm{Volume}(\tau)}$ Relative volume for maturities $\tau \in (11, 30]$	0.094
α Preferred habitat demand slope	5.21	Estimate in KVJ (2012)	−0.746

(a) Table I: calibration parameters.

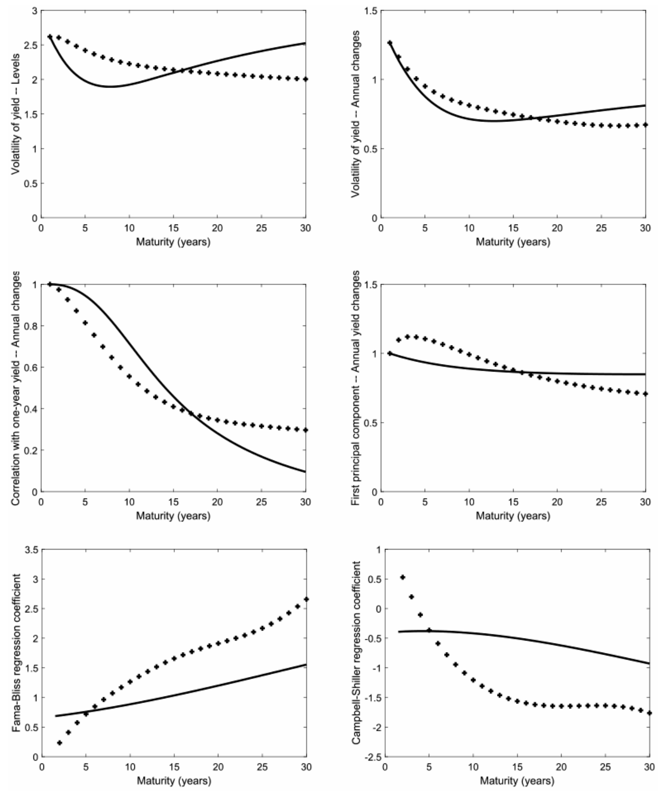

FIGURE 1.—Model-generated and empirical moments for the main sample of nominal yields.

(b) Figure 1: model-generated and empirical moments.

6.2 Figure 2: Effect of forward guidance and Figure 3: Effect of QE

Read: Figure 2 shows the effect of forward-guidance announcements about future short rates. The dashed expectations-hypothesis line moves more than the solid model line, especially at long maturities. Figure 3 shows the effect of QE purchases by maturity: purchases of long bonds have larger effects on long yields, while shorter-maturity purchases peak at shorter maturities.

Interpretation: Forward guidance works mainly through expected short rates and is weakened by arbitrageur risk-bearing constraints. QE works through preferred-habitat demand and can be more targeted, especially when purchases are concentrated at the long end.

Economic message: The comparison clarifies why central banks may use balance-sheet policies when short-rate guidance is insufficient: long-maturity asset purchases directly change the risks that arbitrageurs must absorb.

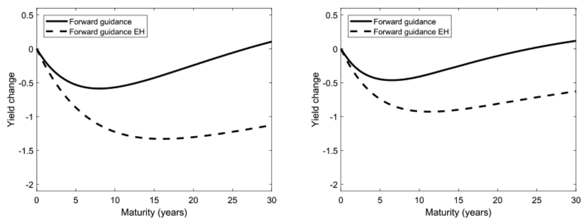

FIGURE 2.—Effect of a forward-guidance announcement about the path of short rates for the calibration based on the main sample of nominal yields.

(a) Figure 2: forward-guidance effect.

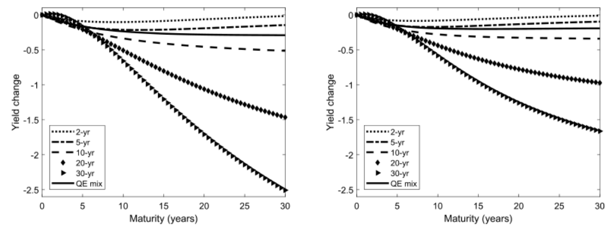

FIGURE 3.—Effect of QE, for the calibration based on the main sample of nominal yields.

$\int_0^\infty \Delta\theta_0(\tau)\,d\tau = -0.12$, that is, QE purchases are 12% of GDP. This is approximately the

(b) Figure 3: QE effect.

7 Interpretive synthesis

7.1 Read

The paper is best read as a bridge between preferred-habitat market narratives and affine term-structure asset pricing. It takes seriously both segmentation and arbitrage. Segmentation matters because clientele demand is maturity-specific. Arbitrage matters because yields cannot move independently at nearby maturities. Limited risk-bearing makes both true at once.

7.2 What the calibration teaches

The calibration teaches that forward guidance and QE operate through different margins. Forward guidance changes the expected path of short rates; QE changes the maturity distribution of risks that arbitrageurs must hold. Because arbitrageurs are risk averse, the latter channel can be powerful even in a no-arbitrage model.

7.3 Why demand risk matters

Without demand risk, preferred-habitat demand effects are global. Demand at any maturity changes the same scalar risk exposure. With demand risk, each demand factor has its own maturity profile, so effects become localized. This distinction is crucial for understanding why central-bank purchases concentrated at long maturities can have especially large long-yield effects.

8 Short summary

- Preferred-habitat investors demand specific maturities.
- Arbitrageurs connect maturity segments but are risk averse.
- Short-rate shocks move long rates through arbitrageur carry trades.
- Bond risk premia are positively related to the term-structure slope.
- In the one-factor model, demand effects are global.
- With stochastic demand factors, demand effects become localized.
- The model is calibrated to U.S. yield moments, trading-volume maturity shares, and demand-elasticity estimates.
- Forward guidance has limited power over long rates because long yields underreact to expected short rates.
- QE can move long rates more effectively when purchases are concentrated at long maturities.

- The paper's incremental contribution is to make preferred habitat a tractable, calibrated, no-arbitrage model for policy and risk-premium analysis.

9 Conclusion

9.1 Core conclusion

Vayanos and Vila formalize the preferred-habitat view of the term structure. They show that limited arbitrage allows maturity-specific demand to affect yields without violating no-arbitrage. Short-rate shocks, demand shocks, bond risk premia, forward guidance, and QE can all be studied in one equilibrium framework.

9.2 Economic intuition

Arbitrageurs are the transmission mechanism. When short rates or preferred-habitat demand shift, arbitrageurs adjust positions across the curve. Because those positions expose them to risk, yields must move to compensate them. This is why preferred-habitat demand can affect prices and why monetary policy transmission is imperfect.

9.3 Incremental contribution in one sentence

The paper's key incremental contribution is to turn preferred habitat from a descriptive idea about maturity clienteles into a calibrated no-arbitrage term-structure model that explains risk premia and quantifies the relative power of forward guidance and QE.

9.4 Final takeaway

The term structure is neither fully segmented nor fully frictionless. It is connected by arbitrageurs whose risk-bearing capacity is limited. This middle ground explains why central-bank asset purchases can move long rates and why forward guidance transmits imperfectly to the long end of the curve.